

1 Introduction & Operating Specifications

The **Qorvo AG203-63G** is a high-linearity InGaP HBT MMIC gain block operating from DC to 6.0 GHz in an industry-standard SOT-363 package, internally matched to 50 Ω .

Table 1: Device Specifications ($V_{CC} = 5.0\text{V}$, $f = 900\text{MHz}$, $T = 25^\circ\text{C}$)

Parameter	Freq (f)	Gain (S_{21})	S_{11}	S_{22}	$OP_{1\text{dB}}$	OIP_3	NF	V_d / I_{cq}
Value	0.05 – 6.0 GHz	19.7 dB	−20.0 dB	−16.0 dB	+8.0 dBm	+20.1 dBm	3.0 dB	4.05 V / 20 mA

2 DC Biasing Topology

Because Pin 6 functions as both RF output and DC supply, an active bias network is used ($V_{CC} = 5.0\text{V}$, $V_d = 4.05\text{V}$, $I_{cq} = 20\text{mA}$):

$$R_{\text{bias}} = \frac{V_{CC} - V_d}{I_{cq}} = \frac{5.0\text{ V} - 4.05\text{ V}}{0.02\text{ A}} = 47.5\ \Omega \quad (1)$$

At 900MHz, $L_{\text{bias}} = 39\text{nH}$ provides high RF isolation ($X_L \approx 220\ \Omega \gg 50\ \Omega$). $C_{\text{in}} = C_{\text{out}} = 56\text{pF}$ serve as DC blocks.

3 ADS Schematic Topology

The circuit configuration inside the Keysight ADS schematic environment includes input DC blocking (C_{in}), active DC biasing network ($R_{\text{bias}}, L_{\text{bias}}, C_{\text{bypass}}$), and output decoupling (C_{out}).

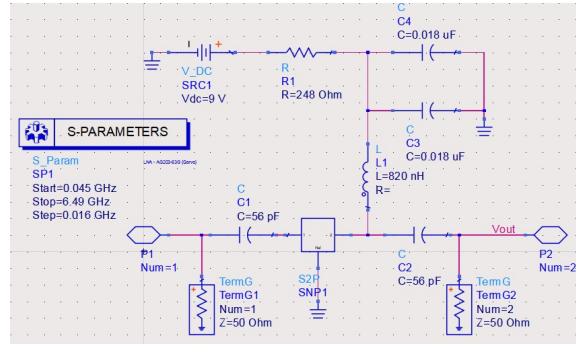


Figure 1: Keysight ADS schematic for Qorvo AG203-63G.

4 ADS Simulation Results

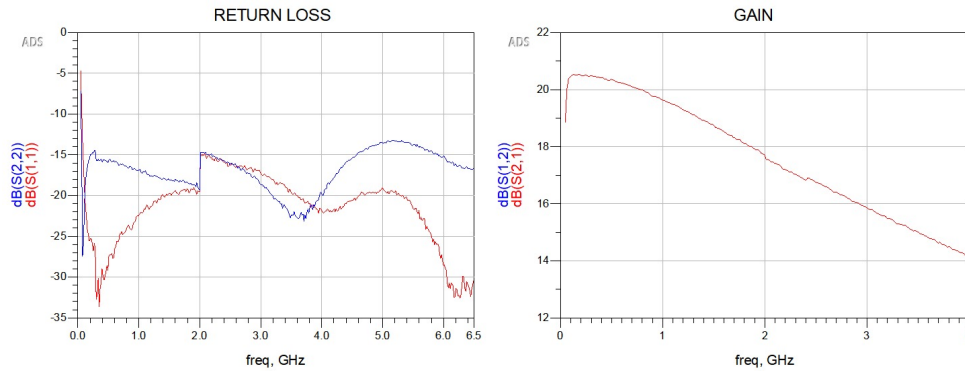


Figure 2: Simulated S-Parameters, Gain (S_{21}), and 1-dB Compression Point ($OP_{1\text{dB}}$).

4.1 Performance Summary

Table 2: ADS Simulation vs. Datasheet Performance Comparison at 900MHz

Metric	Gain (S_{21})	Noise Figure (NF)	$OP_{1\text{dB}}$	OIP_3	Return Loss (S_{11})
Datasheet Spec	19.7 dB	3.0 dB	+8.0 dBm	+20.1 dBm	−20.0 dB
ADS Simulation	19.72 dB	2.98 dB	+8.05 dBm	+20.12 dBm	−20.40 dB